

Reshaping the Teacher's Role in an Age of AI Empowerment: Emotional Leadership and Scaffolding in Generative AI–Integrated Junior High School Instruction

Yeh, Jung-Chen^{1*}

¹Master of Education Administration, College of Humanities and Social Sciences,
Southern Taiwan University of Science and Technology

Email: zoeyeh0822@gmail.com

Abstract

As generative artificial intelligence (GenAI) enters classrooms at scale, technology is reshaping not only the medium through which knowledge is transmitted but also the very nature of teacher–student interaction. In a junior high school information technology course, the researcher observed that students typically approached AI tools with eager anticipation, yet their initial enthusiasm rapidly gave way to frustration and diminished motivation when limited linguistic proficiency prevented them from formulating precise prompts and produced outputs that fell short of their expectations. This micro-level observation reveals that, beneath the rhetoric of "AI empowerment," the mere introduction of technology does not translate naturally into effective learning; rather, the teacher's emotional leadership emerges as the key to helping students cross the digital threshold. Given that current teacher preparation programs concentrate heavily on the technical operation of AI while paying scant attention to the teacher's facilitative role in human–AI collaboration, redefining the teacher's professional identity in the AI era has become a pressing concern for the field—and is the focus of the present study.

This study moves beyond a technology-driven framework to develop a student-centered, teacher-facilitated instructional model that integrates technological mediation with humanistic care, and it examines how teachers deploy emotional leadership and cognitive scaffolding to help students overcome the challenge of "concept visualization" in human–AI co-creation. Adopting an action research design informed by the ADDIE model, the researcher implemented a six-week teaching cycle in a public junior high school in southern Taiwan, with AI positioned as a co-creative partner. Multimodal data—classroom observations, student and teacher reflective journals, and expert feedback—were systematically collected and analyzed qualitatively through triangulation and the constant comparative method. The process foregrounded dynamic, in-situ reflection, with the teacher's facilitation strategies progressively refined across iterative cycles of practice.

Three findings emerged. First, in scaffolding cognition, differentiated "prompt scaffolds"—such as visual cue cards and drop-down menus tailored to individual learner needs—effectively reduced cognitive load, allowing students to bridge linguistic gaps and concentrate on creative thinking. Second, in exercising emotional leadership, when students encountered generative setbacks, teachers' timely empathic responses and reframing prompts transformed anxiety into motivation, ushered students into a state of flow, and elicited an enthusiasm for self-directed learning that exceeded curricular requirements. Third, the study identifies a threefold transformation in the teacher's role in the AI era: from the traditional knowledge

transmitter to a cognitive scaffolder, an ethical gatekeeper, and an emotional leader. These findings offer a practical blueprint for in-service teachers and demonstrate that thoughtful guidance can meaningfully narrow the digital divide. The study thus provides a grounded reference for teacher education institutions seeking to reimagine professional preparation for the AI era, affirming that, however rapidly technology evolves, the essence of education must ultimately return to the empowerment of, and care for, the human person. Drawing on this Taiwan-based practice, the study aims to contribute a perspective on teacher professional development in the AI era that integrates theoretical insight with practical relevance.

Keywords: generative artificial intelligence, teacher professional development, emotional leadership, instructional scaffolding, action research